Paawan Desai

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EDUCATION

The University of Texas at Austin, Cockrell School of Engineering — **Austin, TX**
Bachelor of Science in Electrical and Computer Engineering, Minor in Entrepreneurship — *May 2025*

WORK EXPERIENCE

Gauntlet AI — **Austin, TX**
AI Engineering Fellow – Cohort 7 Challenger (selected from 2,000+ applicants, ~2% admit) — *September 2026 – Present*

- Fine-tuned Qwen3-4B (LoRA) to write exact 94-character NACHA ACH files from plain-language specs: 99/100 exact on holdout (base 0/50), 18/20 on the hardest specs vs. 17/20 for Claude Opus 5.5; on Hugging Face
- Built a clinical co-pilot in OpenEMR that drafts pre-visit briefs from FHIR records and lab PDFs, checking every claim in code; cut report latency from ~90 s to 7–8 s; a 98-case eval gate blocked 22/22 planted regressions
- Built an adversarial eval platform that attacks the deployed co-pilot with code-first oracles (10/11 verdicts decided without an LLM), confirming 3 vulnerabilities, including 2 critical cross-patient record leaks
- Shipped a call-center agent (Claude Haiku 5.5, 7 narrow tools, pgvector RAG) that routed 31/31 calls needing a human to a person and held 46/46 attacks; chose Haiku at ~$0.04 per 100 calls vs. $1.36 for Opus after a cost/quality comparison
- Built a Go aggregator on Cloud Run that matches equivalent Kalshi and Polymarket contracts at 1.00 precision on 95 human-confirmed labels and back-tested 208,064 cross-venue pairs, with 29/29 payout verdicts agreeing
- Also shipped a Monte Carlo pricer on AWS Lambda (1M paths in 0.58 s), Solana markets in Rust and an LA zoning engine

Handshake AI, micro1, Mercor
AI Training & Evaluation Engineer (Contract) — *November 2025 – Present*

- Built LTspice, ngspice, KiCad and Qucs simulation environments that generate LLM training data for circuit design (Handshake AI)
- Built automated tooling and test frameworks that programmatically verify enterprise AI models' API behavior (Mercor)
- Designed prompts and execution environments to test AI-generated code for correctness and edge cases (micro1)

SkyTerra Group
AI & Data Analyst — *April 2025 – November 2025*

- Trained ML models to detect cardiac abnormalities from ECG data on a HIPAA/GDPR-compliant edge-to-cloud stack
- Led field integration of an off-grid hybrid microgrid: battery systems, inverters, generators and remote telemetry

Fundare Finance
Co-Founder & Engineer — *May 2024 – December 2024*

- Led development of a Python and SQL dashboard that tracks revenue growth for small businesses

PROJECTS AND RESEARCH

AI Agent Red Team Engine + AuditLens — github.com/paawandesai/redteam-engine
Adversarial security research (open source) — *April 2026*

- Built an engine that runs RAG-injection, tool-misuse and multi-turn attacks on LangGraph agents and grades the tools each agent actually called; 6 of 10 chains broke all 4 frontier models tested; hardening cut RAG exploits from 14/71 to 6/71
- Mapped 113 prompts to OWASP LLM Top 10, MITRE ATLAS and CWE; AuditLens turns findings into EU AI Act assessments

Hardware Accelerator for Post-Quantum Cryptography (ML-DSA)
Senior design project — *September 2024 – May 2025*

- Led a Verilog ML-DSA accelerator, designing the NTT and random-bit-generator modules and its C driver

Energy AI Hackathon – 1st Place
Predictive energy modeling — *January 2025*

- Forecast grid, diesel and CNG use for fracking with XGBoost/Random Forest and uncertainty modeling (R^2 0.99/0.98/0.97)

SKILLS

Languages: Python, TypeScript, Go, Rust, C/C++, C#, SQL, Verilog
AI & LLM: Claude/OpenAI APIs, LangGraph, RAG (pgvector, ChromaDB), LoRA, vLLM, Hugging Face, Langfuse, LLM-as-judge evals
Infrastructure: AWS (Lambda, CDK, Terraform), GCP Cloud Run, Azure, Modal, Docker, Postgres, Kafka
ML & Hardware: PyTorch, TensorFlow, XGBoost, scikit-learn, time-series forecasting, Edge AI, LTspice, KiCad, microgrids
Honors: AIME Qualifier; Top 2.5% AMC; 3rd Place Nationally, ARML; Captain, Punjabbawockeez